

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. EE Sem - VI Examination, May 2013
EE - 307 Electrical Power System -III

Date: 01/05/2013, Wednesday

Time: 10.00 a.m. to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q-1
1. The initiation of electric arc at the instant of contact separation is caused by 7
(a) thermionic emission of electrons. (b) field emission of electrons.
(c) (a) and (b) both. (d) none of the above.
 2. For magnetic blow out of arc the magnetic field is produced
(a) in the load circuit. (b) at right angles to the axis of the arc.
(c) in line with the axis of the arc. (d) any of the above.
 3. The resistance of an electric arc can be increased by
(a) increasing the concentration of ionized particles.
(b) reducing the arc length,
(c) splitting the arc. (d) increasing the arc x-section.
 4. The arc voltage in a circuit breaker is
(a) in the phase with the arc current. (b) lagging the arc current by 90° .
(c) leading the arc current by 90° . (d) lagging the arc current by 180° .
 5. In a circuit breaker, the active recovery voltage depends upon
(a) circuit conditions. (b) power factor.
(c) armature reaction. (d) all of these.
 6. The RRRV depends upon the
(a) type of the circuit breaker.
(b) capacitance of the system only.
(c) inductance of the system only.
(d) inductance and capacitance of the system.
 7. In air-blast circuit breaker resistance switching is used to
(a) reduce the magnitude of fault current.
(b) control the CB operating time.
(c) damp out the fast transient.
(d) change the fault current power factor.
- Q-2
- A. With neat diagrams explain the term "Current Chopping". Also find out voltage across contact of the circuit breaker when it interrupts the instantaneous value of magnetising current of 2.93 amp. The inductive reactance of the transformer is 31700 ohm, and the capacitance between phase and ground is 5000pF. Take frequency as 50 Hz. 6
- B. Explain the arc interruption theories. 4
- C. With neat diagrams explain the phenomena of "First pole to clear factor" for L-L-L fault and L-L fault 4

OR

- Q-2 A. With necessary diagram explain the terms (i) Transient recovery voltage, 6
(ii) active recovery voltage and (iii) recovery voltage. Also explain the effect of
(a) power factor and (b) armature reaction on Transient recovery voltage.
- B. In a short circuit test on a 3 pole, 110 KV circuit breaker, power factor of fault 6
was 0.4; the recovery voltage was 95% of full line value. The breaking current
was symmetrical. The frequency of oscillations of restriking voltage was 15000
c/s. Estimate the average rate of rise of restriking voltage. The neutral is grounded
and fault involves earth. Neglect the first pole to clear factor.
- C. Give reason: For circuit breaker the value of rated short circuit making current is 2
always higher than rated short circuit breaking current.
- Q-3 A. Explain the major differences between "High resistance interruption" and "Low 3
resistance interruption".
- B. Write short note on selection of circuit breaker. 4
- C. With neat sketch explain the construction and working of Air Break Circuit 7
Breaker.

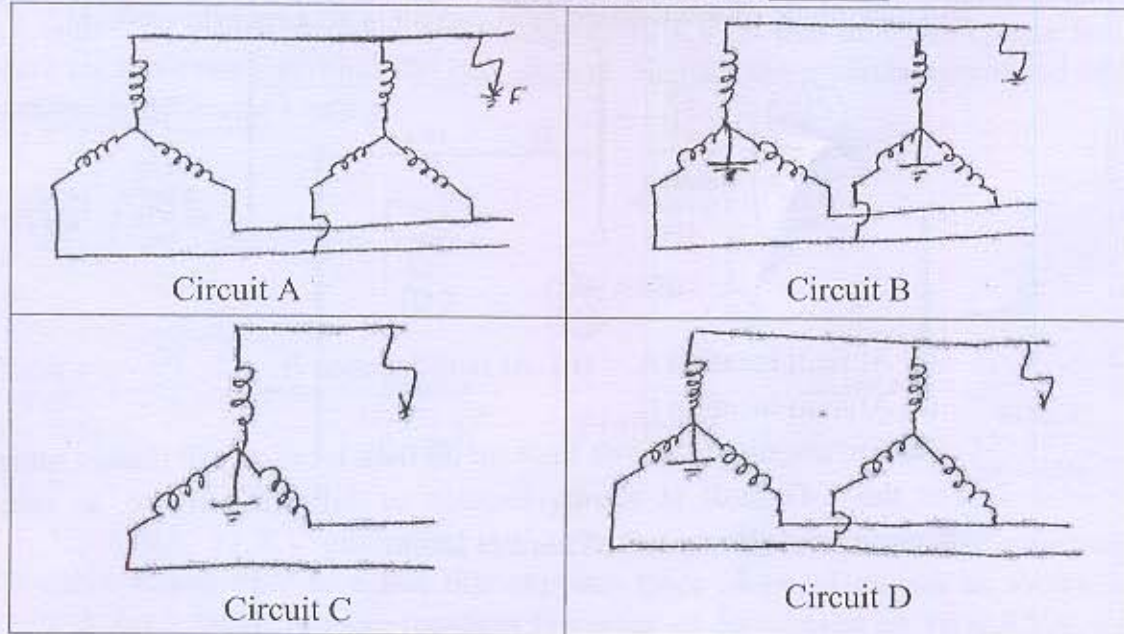
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- Q-3 A. With neat diagram explain the resistance switching in Air Blast Circuit Breaker. 7
- B. Explain the construction and working of Vacuum Circuit Breaker. 7

SECTION – II

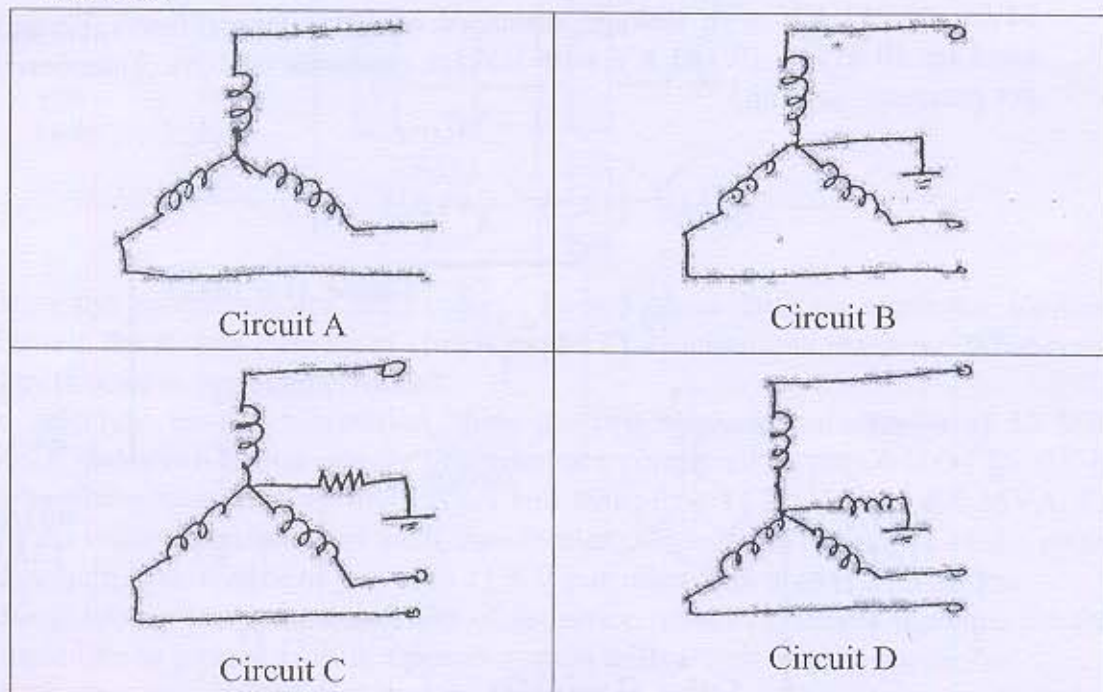
- Q-4 1. If we interconnect two substation by a reactor which has higher pu value. The 7
new positive sequence Thevenin reactance value will _____ the old (Without
interconnection) positive sequence Thevenin reactance value at particular point.
(a) More than
(b) Less than
(c) Equal to
(d) More or less depend on condition.
2. The power invariance theorem says that...
(a) In all sequence network power remains same.
(b) zero sequence network doesn't consume any power.
(c) Power consume as per phase quantity is three times power consumes as per
positive sequence quantity.
(d) Power consume as per phase quantity is equal to sum of the power consume
in positive, negative and zero sequence quantity.
3. For a particular fault, $I_{c0}=0$ and $I_{b2}=j6.88$ pu. The possible fault is ____
(a) L-G (b) L-L (c) L-L-G (d) L-L-L

4. LG fault value is maximum at shown location for the _____.



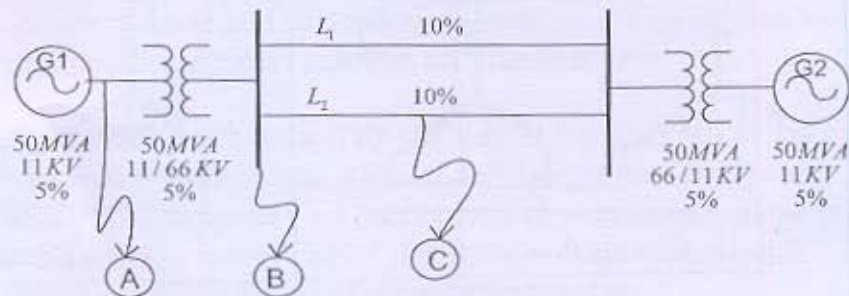
- (a) Circuit A
- (b) Circuit B
- (c) Circuit C
- (d) Circuit D
- (e) All the circuit has same LG fault current.

5. Which circuit has the maximum fault current for L-L-L fault at generator terminal?

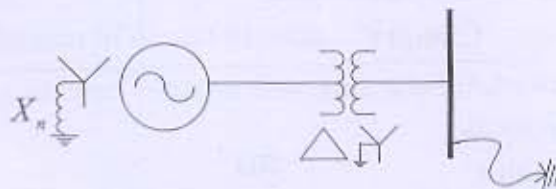


- (a) Circuit A (b) Circuit B
- (c) Circuit C
- (d) Circuit D
- (e) All the circuit has same LG fault current.

6. For the system shown in the diagram given below, at which fault location per unit magnitude of three phase fault current is maximum?

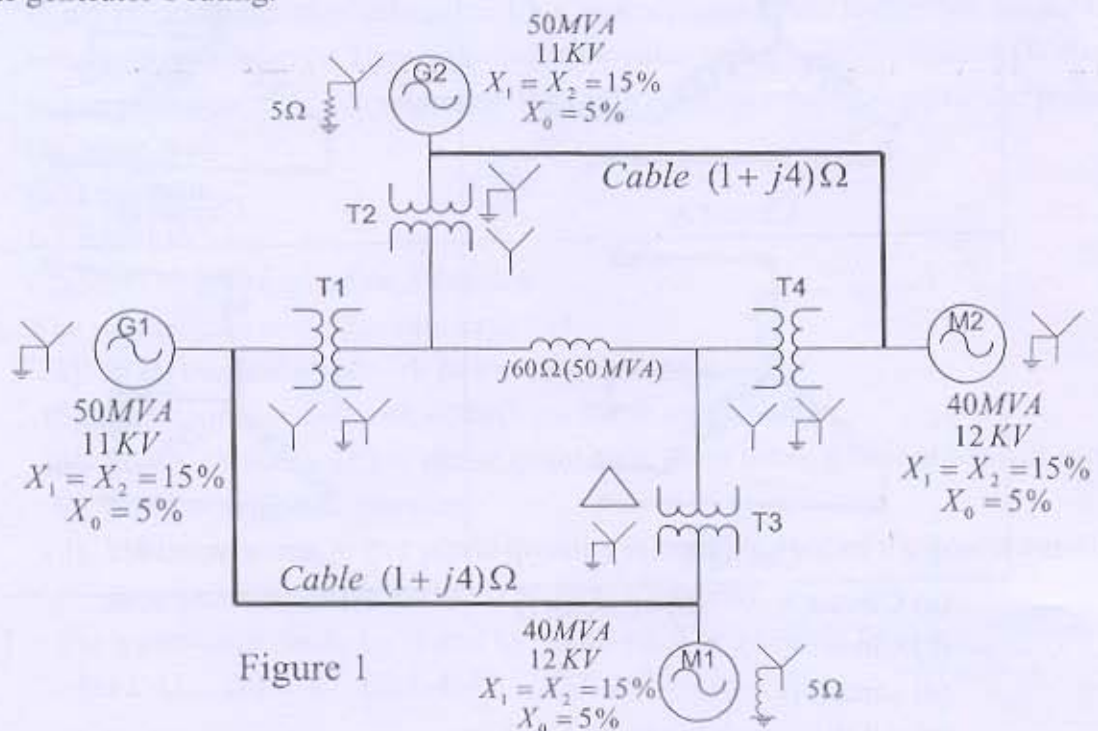


- (a) At fault location A. (b) At fault location B.
 (c) At fault location C.
 (d) p.u. magnitude of 3- Φ fault for all fault location will remain same.
7. For the L-G fault at shown location of following figure, as value of X_n increases, the effect on fault current is.....

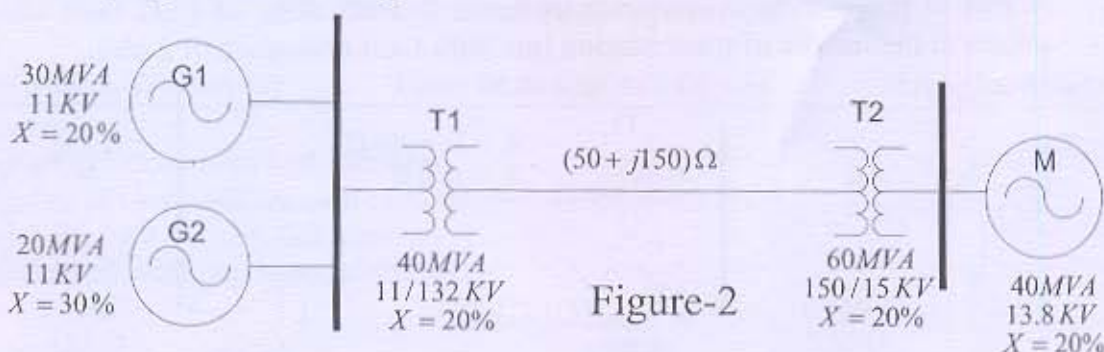


- (a) Fault current increases (b) Fault current decrease
 (c) Fault current remains same (d) No fault current will flow

- Q-5 (A) Draw the p.u. reactance diagram for the part of the power system shown in the figure 1. Take base as per generator 1. Transformers T1 and T3 are rated for 10 MVA, 12/132 KV with leakage reactance of 10%. Transformers T2 and T4 are rated for 10 MVA, 13/143 KV with leakage reactance of 10%. Take base value as per generator 1 rating.

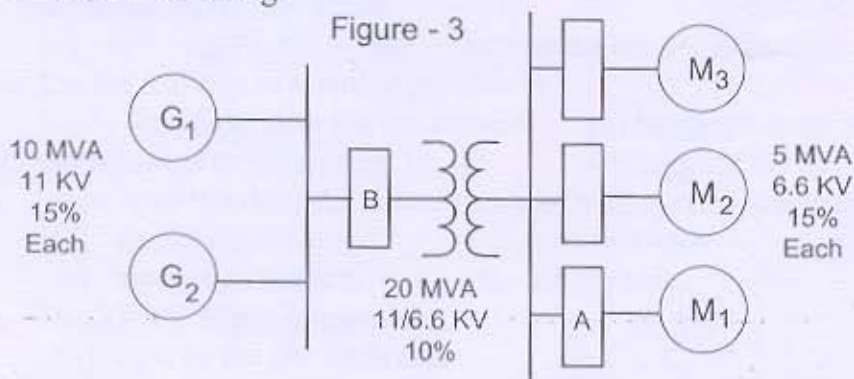


- (B) A simple power system is shown in figure 2. When the motor was running at unity p.f. with 80% load with motor bus voltage of 12 KV, at that time three phase fault occur on generator bus. Find the fault current. Neglect the prefault current and take base as per generator 1 rating.



OR

- Q-5 (A) Two 10 MVA, 11 KV generators with 15% sub-transient reactance is connected through a transformer to a bus that supplies three identical motors as shown in figure 3. Each motor has sub-transient reactance of 15% based on 5000 KVA, 6.6 KV. Three phase rating of transformer is 20 MVA, 11/6.6 KV and leakage reactance is 10%. Bus voltage is 6.6KV when a 3-Φ fault occurs at the terminal of one motor as shown in figure. Neglecting the prefault current, find out the current through breaker A and current through breaker B during fault. Take base MVA as per transformer rating.



- (B) Draw the waveforms for fault current for a 3-phase fault on alternator terminals. Explain the steady state short circuit model of synchronous machine. What is their significance in fault calculations?
- (C) In "changa" 66 KV sub-station, there are two identical transformer of 50 MVA, 66KV delta/11KV Star solidly grounded are connected in parallel. On 66 KV bus three phase fault level is 1000 MVA and that of on 11 KV bus is 400 MVA. Find the pu leakage reactance of each transformer. Neglect the resistance of the system, also there is no source of power to 11 KV bus other than above information.
- Q-6 (A) Using appropriate interconnection of sequence networks, derive the equation for a single line to ground fault in a power system with a fault impedance of Z_f .
- (B) Draw the zero sequence network diagram of figure 1(Drawn on page no.4 for question 5(A)). Negative sequence impedance of cable is 100% of the positive sequence reactance of that cable. Zero sequence impedance of cable is 300% of the positive sequence reactance of that cable. The construction of the reactor is same as the transformer. Take base value as per generator 1 rating.

OR

- Q-6 (A) Using appropriate interconnection of sequence networks, derive the equation for a line to line fault in a power system with a fault impedance of Z_f .
- (B) A part of power system is shown in figure 4. Find value of LLG fault when fault occurs at the middle of transmission line with fault resistance of 2 ohm.

